



Assembly Instructions

Gas engine-generator set – Power container
Series 400

M060802/02E

Product name	Model number	Engine model	Application group
MTU 12V400 GS	GG12V0400A1	B3042Z7 E3042Z6 B3042L8 B3042L9 E3042L8 E3042L9 E3042D4	3A, continuous operation, unrestricted
MTU 12V400 GS	GG12V0400C1	B3042Z7 E3042Z6 B3042L8 B3042L9 E3042L9 E3042D4	3A, continuous operation, unrestricted
MTU 6R400 GS	GG06R400A1	B3066Z8 E3066Z5 B3066L8 B3066L9 E3066L8 E3066L9	3A, continuous operation, unrestricted
MTU 6R400 GS	GG06R400A2	E3066D4	3A, continuous operation, unrestricted
MTU 6R400 GS	GG06R400C1	B3066Z8 E3066Z5 B3066L8 B3066L9 E3066L8 E3066L9	3A, continuous operation, unrestricted

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1 Preface

1.1 Preface

These Assembly Instructions only apply to products of the Series 400 and Series 4000 CHP containers by MTU.

The instructions, notes and warnings in this document must be observed in all circumstances to guarantee correct installation.

This description is based on the following items that have to be carried out prior to putting into operation:

- Assembly of roof-mounted equipment
- Removal of transport covers / securing elements
- Establishment of hydraulic connection
- Establishment of electrical connection

The container is transported without external attachments. These are usually delivered in a separate transport container.

The small parts required for re-installation (screws, gaskets, etc.) are stowed in the CHP container for transport.

The following accompanying documents must be complied with, in addition to the Assembly Instructions:

- Planning drawing
- Schematic diagrams
- TEN data
- Arrangement drawings
- Operating Instructions (see Business Portal)
- Maintenance Schedule (see Business Portal)
- Fluids and Lubricants Specifications (see Business Portal)
- MTU standards

The relevant documents are defined in the specifications related to the order concerned.

Before starting work, the delivery must be checked for completeness and intactness.

The Assembly Instructions do not relieve those responsible for the system from independently ensuring that installation is carried out in an appropriate manner.

The object of delivery from MTU complies with the EC Machinery Directive and German specifications and standards.

When the object of delivery is used abroad or in other special regions, MTU is not responsible for the observance of statutory and other specifications / standards applicable at the place of use.

Local illustrations may deviate from the images in this description because options that were not selected by the customer have been faded out. Nor do these illustrations contain any customer adaptations.

Definition of MTU

MTU refers to Rolls-Royce Power Systems AG and MTU Friedrichshafen GmbH or an affiliated company pursuant to Section §15 AktG (German Stock Corporation Act) or a controlled company (joint venture).

2 Safety

2.1 Gas systems – Intended use

Intended use

The machine is intended to be used exclusively for electrical power generation and waste heat utilization in stationary operation, either in sufficiently ventilated rooms or in container solutions.

A gas/air mixture is fed to the internal combustion reciprocating piston engine. This is converted into mechanical work in the individual cylinders through combustion. The flywheel of the engine drives the synchronous generator connected via an elastic coupling, resulting in a conversion to electric energy. The engine and the generator are joined to the base frame with resilient, vibration-damped elements. For mains frequency 60 Hz, depending on the series either a gearbox is used between the engine and generator (Series 4000) or the engine speed is increased accordingly (Series 400). The waste heat generated during the combustion process is dissipated to the surroundings via coolers (power generator) and/or made available to the customer via heat exchangers (combined heat and power plant).

Area of application	Applicable	Comment
Industry	Yes	–
Business/Service	Yes	In some individual cases, however, such as with swimming pool facilities, schools, heating stations for apartment buildings and the like, no end user in the classic sense is involved
Household	No	Operation is excluded
Potentially explosive area	No	Operation is not permitted in potentially explosive areas
Underground	No	Operation is excluded
Mobile use	No	Operation is excluded

Table 1: Area of application of machine

There are 3 operating modes of the machine:

- Mains parallel operation
- Mains failure operation
- Isolated operation

Foreseeable misuse

Any other use, particularly misuse, is considered as being contrary to the intended purpose. Such improper use increases the risk of injury and damage when working with the product. The manufacturer shall not be held liable for any damage resulting from improper use.

Operation

Operation is not permissible if the physical conditions specified on the technical data sheet or on/in other documents (concerning temperature, pressure, site altitude, etc.) are not complied with. Operation is not permissible without protective equipment, usage of approved spare parts, exclusion of conversions, warning of foreseeable misuse.

Operation is only permissible with the use of fluids and lubricants (natural gas, biogas, quality-compliant gas, quality-compliant water, lubricants) in accordance with the (→ Fluids and Lubricants Specifications).

Operation is only permissible with preservation agents approved by MTU in accordance with the (→ Preservation and Represervation Specifications).

The specifications of the manufacturer will be amended or supplemented as necessary. Prior to operation, make sure that the latest version of these are adopted. The latest version can be found at:

- <http://www.mtu-solutions.com>

Surroundings, climate, temperature

The weather plays no role with regard to the intended use of the engine-generator set/module provided that operation takes place inside an installation room. For container applications, however, the permissible outdoor temperatures at the installation site play a decisive role. The product-specific ambient conditions such as climate and temperature are indicated in the technical description.

Prerequisite for proper use

Fulfillment of all the requirements set out in the "Installation conditions" document.

Any use that does not fall within the described limits for use is regarded as improper.

Work which requires communication with the control units must only be carried out with a diagnosis software or with tools approved by the manufacturer.

Modifications or conversions

Unauthorized changes to the product breach the conditions concerning its intended use and compromise safety.

Only modifications or conversions expressly authorized by the manufacturer are regarded as compliant in this sense. The manufacturer shall not be held liable for any damage resulting from unauthorized modifications or conversions.

2.2 Important provisions for all products

General

This product may pose a risk of injury or damage in the following cases:

- Incorrect use
- Operation, maintenance and repair by unqualified personnel
- Changes or modifications which are neither made nor authorized by the manufacturer
- Noncompliance with the safety instructions and warning notices

Nameplates

The product is identified by nameplate, model designation or serial number. This data must match the specifications in these instructions.

Nameplates, model designation or serial number can be found on the product.

All EU-certified engines delivered by MTU come with a second nameplate. This second nameplate is delivered "loosely" with the engine. If the nameplate secured to the engine after installation in the vehicle/system is not visible without the removal of components, the system integrator must install the second nameplate in a clearly visible area on the vehicle/system.

Emissions regulations and emissions label

Responsibility for compliance with emissions regulations

Modification or removal of any mechanical/electronic components or the installation of additional components including the execution of calibration processes that might affect the emissions characteristics of the product are prohibited by emissions regulations. Emissions-related components must only be serviced, exchanged or repaired if the components used for this purpose are approved by the manufacturer.

Noncompliance with these specifications will invalidate the design type approval or certification issued by the emissions regulation authorities. The manufacturer does not accept any liability for violations of the emissions regulations.

The product must be operated over its entire life cycle according to the conditions defined as "Intended use" (→ Page 6).

Emissions certification applicable to engines with EPA Nonroad Tier 4 emissions certification in accordance with 40 CFR 1039 and with EPA Locomotive emissions certification in accordance with 40 CFR 1033

Extract from the standard:

Failing to follow these instructions when installing a certified engine in a piece of nonroad equipment violates federal law (40 CFR 1068.105(b)), subject to fines or other penalties as described in the Clean Air Act.

Extract from the standard:

If you install the engine in a way that makes the engine's emission control information label hard to read during normal engine maintenance, you must place a duplicate label on the equipment, as described in 40 CFR 1068.105.

When fitting the second label, the requirements of 40 CFR 1068.105(c) must be followed and observed. This paragraph describes the process for requesting and fitting the label, the documentation obligations and storage obligations for the required documents.

Replacing components with emissions labels

On all engines from MTU fitted with emissions labels, these labels must remain on the engine throughout its operational life.

Exception: Engines used exclusively in land-based, military applications other than by US government agencies.

Please note the following when replacing components with emissions labels:

- The relevant emissions labels must be affixed to the spare part.
- Emissions labels shall not be transferred from the replaced part to the spare part.
- The emissions labels must be removed from the replaced part and destroyed.

2.3 Personnel and organizational requirements

Organizational measures of the user/manufacturer

This manual must be issued to all personnel involved in operation, maintenance, repair, assembly, installation, or transportation.

Keep this manual handy in the vicinity of the product such that it is accessible to operating, maintenance, repair, assembly, installation, and transport personnel at all times.

Personnel must receive instruction on product handling and repair based on this manual. In particular, personnel must have read and understood the safety requirements and warnings before starting work.

This is important in the case of personnel who only occasionally perform work on or around the product. Such personnel must be instructed repeatedly.

Personnel requirements

All work on the product must be carried out by trained, instructed and qualified personnel only:

- Training at the Training Center of the manufacturer
- Qualified personnel from the areas mechanical engineering, plant construction, and electrical engineering and also for work with live parts

The operator must define the responsibilities of the personnel involved in operation, maintenance, repair, assembly, installation, and transport in writing.

Personnel shall not report for duty under the influence of alcohol, drugs or strong medication.

Clothing and personal protective equipment

Always wear appropriate personal protective equipment, e.g. safety shoes, ear protectors, protective gloves, goggles, breathing mask. Follow the instructions concerning personal protective equipment in the respective descriptions of the individual activities or the manufacturer's documentation included in the delivery.

When working on live components, in particular on electrical switchgear or on high-voltage component, wear special protective clothing according to IEC 61482.

Safe handling of Substances of Very High Concern pursuant to the REACH regulation (Registration, Evaluation, Authorization and restriction of CHemicals): We recommend wearing protective gloves at all times in order to reduce risk when working.